Green Haven Foods is an FMCG company handling perishable snack products with a short shelf life.

The company faces:
Poor demand forecasting
Delivery delays
Product waste
Manual operations
Inventory planning problems

The dataset analysis was performed to identify:
Demand trends
Inventory shortages
Stockout patterns
SKU-wise sales behaviour
Supply chain inefficiencies

Month	SKU	Opening Inventory (Units)	Sales (Units)	Closing Inventory (Units)
Jan	A	10000	7800	2200
Jan	B	5000	4500	500
Jan	C	8000	6400	1600
Feb	A	2200	6000	3800 (Stockout)
Feb	B	500	4200	3700 (Stockout)
Feb	C	1600	5000	3400 (Stockout)
Mar	A	0	9000	9000 (Stockout)
Mar	B	0	4000	4000 (Stockout)
Mar	C	0	5500	5500 (Stockout)
Apr	A	5000	6500	1500 (Stockout)
Apr	B	6000	5800	200
Apr	C	7000	6000	1000
May	A	10000	8800	1200
May	B	7000	6500	500
May	C	8000	6400	1600
Jun	A	7000	6800	200
Jun	B	6000	5500	500
Jun	C	7500	5900	1600

SKU-wise Sales

SKU A
7800 + 6000 + 9000 + 6500 + 8800 + 6800

= 44,900 units
SKU B
4500 + 4200 + 4000 + 5800 + 6500 + 5500
= 30,500 units
SKU C
6400 + 5000 + 5500 + 6000 + 6400 + 5900
= 35,200 units

Total Company Sales
44900 + 30500 + 35200
= 110,600 Unit

SKU A has the highest demand.

SKU	Total sales
A	44900
B	30500
C	35200

Interpretation
SKU A is the best-selling product.
Requires:
Higher safety stock
Better forecasting
Priority replenishment

Average Monthly Sales
SKU A
44900/6
= 7483 units per month

SKU B
30500/6
= 5083 units per month

SKU C
35200/6
= 5867 units per month

Analysis
Demand ranking:
SKU A > SKU C > SKU B
Inventory allocation should follow this order.

Inventory Analysis

January Analysis
SKU A
Opening Inventory = 10000
Sales = 7800
Closing Inventory = 2200
Balanced inventory.

SKU B
Opening = 5000
Sales = 4500
Closing = 500
Very low remaining stock.
Risk of shortage.

SKU C
Opening = 8000
Sales = 6400
Closing = 1600
Moderately balanced.

February Analysis
Multiple stockouts observed.
SKU A:
Opening = 2200
Sales = 6000
Demand exceeded inventory.
Stockout occurred.

SKU B:
Opening = 500
Sales = 4200
Severe shortage.

SKU C:
Opening = 1600
Sales = 5000
Critical shortage.

Stockout Analysis
Major finding

Several months contain stockout remarks.
Stockout Months:

Month	SKU
FEB	A, B, C
MAR	A, B, C
APR	A

Total stockout cases:
7 stockout incidents

Business Impact
Stockouts cause:
Lost sales
Customer dissatisfaction
Delivery disruption
Emergency replenishment cost
Poor service level
Indicates weak forecasting capability.

Demand–Supply Gap Analysis
Demand Gap
=Sales−Opening Inventory
Feb SKU B
4200 − 500
= 3700 units shortage

Mar SKU A
9000 − 0
= 9000 units shortage
Critical planning failure.

Key Finding
March shows a severe supply shortage.
Opening inventory:
0 units across all SKUs.
This indicates:
Procurement delays
Production planning failure
Replenishment breakdown

Trend Analysis

Sales Trend
SKU A:
7800 → 6000 → 9000 → 6500 → 8800 → 6800
Demand fluctuates strongly.
Requires:
AI forecasting implementation

SKU B:
Stable moderate demand.
Inventory planning is easier.

SKU C:
Relatively steady.
Good candidate for optimised replenishment cycle.

Root Cause Analysis
Identified Issues:
1.Poor Demand Forecasting
Evidence:
Repeated stockouts.
2. Weak Inventory Planning
Evidence:
Opening inventory is frequently insufficient.
3. Manual Supply Chain Operations
Creates:
delayed decision making
low visibility
inaccurate planning
4. No Predictive Analytics
The company reacts after a shortage occurs.

Recommended Digital Improvements
AI Forecasting

Use machine learning / Excel forecasting to predict:
monthly demand
transport cost
replenishment needs

ERP Integration

Inventory Optimization
Adopt:
Safety Stock Model.
Safety Stock = Average Demand * Lead Time
Automated Alerts
Trigger alerts when inventory drops below the threshold.

Conclusion
The data analysis identified major operational weaknesses.
Key findings:
Total sales = **110,600 units**

Highest demand
SKU A
Multiple stockout events detected.
March experienced critical shortages.
Forecasting and inventory planning are inadequate.
Digital forecasting and ERP integration are required.

Improvements expected after digitisation:
Fewer stockouts
Increased service levels
Less waste
More satisfied customers
Increased operational efficiency

Architecture planning

Existing Supply Chain Problem
Green Haven Foods system operates using:

Manual documentation
Disconnected departments
Poor forecasting
Limited visibility
Manual quality checks

Major issues

Problem	Impact
Manual procurement	procurement delays
Weak forecasting	Stockouts
Limited warehouse visibility	Excess waste
Delivery delays	Reduced freshness
Manual quality inspections	Human error

Proposed Digital Supply Chain Workflow

1. Supplier
2. Digital procurement system
3. ERP platform
4. Production planning
5. Smart warehouse
6. IOT logistics tracking
7. Retail delivery
8. Customer

Integrated system architecture

Procurement Management

Existing problem: Procurement is handled manually
This is creating
Delayed ordering
Poor supplier coordination
Inaccurate material planning

Digital solution
Implementation of the Digital procurement system + ERP
Functions
Automated purchase orders
Supplier database management
Lead time tracking

Material requirement planning

Workflow
1. Demand forecast
2. ERP calculates the material requirement
3. Purchase requisition generated
4. Supplier receives digital P.O
5. Material delivery scheduled
Benefits
Faster procurement
Lower paperwork
Better supplier visibility
Reduced stock shortage

Production planning
Existing Problem
Production planning disconnected from the demand.
Consequences:
overproduction
underproduction
waste generation

Digital Solution
Integrate:
ERP + AI Forecasting Module
AI predicts:
monthly sales demand
production requirement
raw material consumption

Workflow
1. Historical Sales Data
2. AI Forecast Model
3. Production Requirement Generated
4. Production Schedule Created

Benefits
Reduced product waste
Better production planning
Higher forecasting accuracy

Warehouse Management
Existing Problem
Warehouse processes mostly manual.
Problems:
inaccurate inventory tracking
delayed stock updates
poor material visibility

Digital Solution
Implement:
Smart Warehouse Management System (WMS)
Components:
RFID tags
Barcode scanning
Inventory dashboard
Automated stock alerts

Workflow
1. Goods Received
2. Barcode / RFID Scan
3. Inventory Automatically Updated
4. Warehouse Dashboard Updated

Benefits
real-time inventory visibility
reduced stockout risk
faster warehouse operations

Logistics & Delivery Integration
Existing Problem
Deliveries to remote locations experience delays.
Problem impacts:
reduced freshness
customer dissatisfaction
spoilage risk

Digital Solution
Deploy:
IoT-Enabled Logistics Monitoring
Tools:
GPS tracking
Temperature sensors
Route optimization software

Workflow
1. Shipment Dispatch
2. GPS Tracking Activated
3. Live Location Monitoring
4. Temperature Monitoring
5. Delivery Confirmation

Benefits
Reduced spoilage
Better shipment visibility
Improved delivery reliability

Customer & Retail Integration
Existing Problem
Low visibility after shipment delivery.
Digital Solution
Implement:
Customer Portal / ERP Integration.
Functions:
delivery tracking
order confirmation
complaint management
product traceability

Workflow
1. Delivery Completed
2. Digital Proof of Delivery
3. Customer Confirmation
4. ERP Database Updated

Benefits
improved customer satisfaction
quicker complaint resolution
stronger service visibility

Proposed digital architecture
1. Suppliers
2. Digital procurement system
3. Erp platform
4. Ai demand forecasting
5. Production planning
6. Smart warehouse (WMS)
7. IOT delivery tracking
8. Retailer / customer
9. Blockchain record update

Expected improvements

KPI	Current	Digital system
Forecast accuracy	Low	High
Delivery visibility	Poor	Real time
Stockouts	Frequent	Reduced
Waste	High	Lower
Response time	Slow	Faster

AI FORECASTING & TRANSPORTATION COST OPTIMIZATION

Objective
The objective of AI forecasting is to use historical sales data and Excel forecasting techniques to:
Predict future demand
Estimate transportation cost
Reduce stockouts
Improve inventory planning
Optimize logistics cost

Existing Business Problem
Green Haven Foods currently faces:
1. inaccurate forecasting
2. delivery delays
3. product waste
4. inefficient transportation planning
Current forecasting method:
Manual estimation

PROBLEM	IMPACT
Poor demand prediction	Stock shortages
Incorrect production planning	Excess inventory
Unplanned deliveries	High transport cost
No predictive analytics	Reactive decision making

AI Forecasting Concept
AI forecasting uses:
1. Historical demand data
2. Trend patterns
3. Predictive models
4. Automated analysis

to estimate future business requirements.

Historical Sales Data Used
Example:
SKU A Monthly Sales:

MONTH	SALES
JAN	7800
FEB	6000
MAR	9000
APR	6500
MAY	8800
JUN	6800

Sales fluctuate significantly.
Therefore, forecasting is necessary.

Example Calculation — SKU A
Historical Sales:
7800
6000
9000
6500
8800
6800

Average Forecast

Calculation:
(7800+6000+9000+6500+8800+6800)/6
= 44,900 /6
= 7483 Units

Expected Next Month Demand:
SKU A
7483 Units

SKU B
Sales:
4500,4200,4000,5800,6500,5500
Forecast:
30500/6
= 5083 Units

SKU C
Sales:
6400,5000,5500,6000,6400,5900
Forecast:
35200/6
= 5867 Units

Forecast summary table

SKU	Forecast demand
SKU A	7483
SKU B	5083
SKU C	5867

Analysis
Highest forecasted demand:
SKU A
Business implication:
1. Priority production scheduling
2. Higher safety stock
3. Higher replenishment planning

Trend Analysis
Demand trend shows fluctuations.
Example:
SKU A:

7800:6000:9000:6500:8800:6800
Observations:
1. irregular demand
2. seasonal variation
3. unstable planning conditions

Without forecasting:
Production becomes inaccurate.

AI can improve transportation planning using:
Route Optimisation: shortest delivery routes.
Benefits:
1. lower fuel usage
2. faster delivery

Load Optimization
AI analyses truck utilisation.
Example:
Truck Capacity: 100%
Actual Usage: 65%
AI recommendation: Combine shipments.

Predictive Scheduling
AI predicts: High demand periods.

Result:
better dispatch planning.
Fuel Consumption Optimisation
AI monitors:
1. driving patterns
2. idle time
3. route efficiency
4. Result:

Reduced logistics expense.

AI Enabled forecast workflow
1. Historical sales data
2. Excel/ AI Forecast model
3. Demand prediction
4. Production planning
5. Transportation planning
6. Cost optimization

Recommended AI tools

TOOL	PURPOSE
Excel forecast sheet	Demand prediction
Power BI	Trend dashboard
SAP Analytics	Supply chain planning
Machine learning models	Advanced forecasting

Expected improvements after AI adoption

KPI	CURRENT	AFTER AI
Forecast accuracy	Low	Higher
Stockouts	Frequent	Reduced
Delivery planning	Reactive	Predictive
Transportation cost	High	Lower
Inventory waste	High	Reduced

DELIVERY TRACKING & WAREHOUSE MONITORING

The objective of IoT (Internet of Things) implementation is to improve:

Delivery visibility

Warehouse monitoring

Product freshness control

Inventory tracking

Real-time operational decision making.

Current issues include:

delivery delays

manual warehouse checks

poor visibility during transport

spoilage risk

inventory tracking inefficiency

Since products are **perishable snacks**, maintaining freshness is critical.

Current supply chain challenges

Problem	Business impact
No live delivery tracking	Late shipments
Manual warehouse monitoring	Human errors
No temperature control visibility	Product spoilage
Poor stock visibility	Inventory accuracy

Delayed issue detection Customer dissatisfaction

IoT Solution for Delivery Tracking
Existing Problem
Deliveries to urban and remote regions experience:
shipment delays
limited visibility
freshness loss
Manual tracking is ineffective.
Proposed Solution
Implement:
GPS-Based Delivery Tracking System
Technology Used:
GPS Sensors
Mobile Connectivity
Cloud Dashboard

Delivery workflow
1. Order dispatch
2. GPS device activated
3. Live vehicle tracking
4. Location data sent to the dashboard
5. Monitoring delivery status
6. Delivery confirmation

Benefits
Real-time shipment visibility
Faster problem detection
Improved delivery planning
Reduced late deliveries
Better customer satisfaction

Temperature Monitoring During Delivery
Existing Problem
Perishable food products require controlled temperature.
Without monitoring:
- quality deterioration occurs
- spoilage risk increases

Proposed Solution
Install:
IoT Temperature Sensors
Sensors continuously measure:

- cargo temperature
- humidity
- storage environment

Delivery monitoring workflow
1. Truck loading
2. Temperature sensor activated
3. Real-time data tracking
4. Data sent to the dashboard
5. Alert generated if the limit is exceeded

Benefits
Product freshness protection
Reduced spoilage
Improved compliance
Better product quality control

IoT Solution for Warehouse Monitoring
Existing Problem: Warehouse monitoring relies on manual inspection.
Problems:
- delayed stock updates
- inventory inaccuracies
- unnoticed environmental changes

Proposed Solution
Implement:
Smart Warehouse Monitoring System
Technology:
RFID Tags
Barcode Scanners
Temperature Sensors
Humidity Sensors
Inventory Dashboard

IOT-integrated supply chain workflow
Supplier
ERP procurement system

Production
Smart warehouse (RFID + Sensors)
GPS delivery tracking (Temperature monitoring)
Customer delivery
Cloud dashboard

Recommended IoT Tools

Tool	Application
GPS tracker	Delivery monitoring
RFID Tags	Inventory tracking
Barcode scanner	Warehouse operations
Temperature sensor	Cold chain monitoring
Humidity sensor	Storage condition control
Cloud dashboard	Data visualization

Expected improvements after IoT adoption

KPI	Current	IoT enabled
Delivery visibility	low	Real time
Inventory accuracy	Moderate	High
Spoilage risk	High	Lower
Warehouse monitoring	Manual	Automated
Delay detection	Slow	Immediate

BLOCKCHAIN PLANNING

Objective

The objective of blockchain implementation is to improve:

1. Product Traceability
2. Compliance Monitoring
3. Delivery Verification
4. Automated Payment Processing
5. Supply Chain Transparency

Existing Supply Chain Problems

- manual compliance verification
- limited shipment traceability
- delayed payment approval
- documentation dependency
- data manipulation risk

Business impact

Problem	impact
Manual records	Data errors
Limited visibility	Poor traceability
Payment delays	Supplier dissatisfaction
Compliance failures	Quality risk
Paper based verification	Slow operations

Supply Chain Application

Blockchain stores:

- supplier information
- production records
- shipment details
- delivery confirmation
- compliance status

Block chain-based supply chain workflow

1. Supplier
2. Raw material dispatch record
3. block chain ledger updated
4. Production batch generated
5. Shipment tracking added
6. Delivery verification completed
7. Smart contract triggered
8. Automatic supplier payment

Blockchain benefits

BENEFIT	IMPACT
Traceability	Higher visibility
Smart contracts	Faster payments
Data security	Lower fraud risks
Compliance automation	Better quality control
transparency	Improved stake holder trust

Expected business improvements

1. Reduced paperwork
2. Faster compliance verification
3. Quicker supplier settlement
4. Improved recall capability
5. Enhanced customer confidence

Blockchain implementation can help Green Haven Foods

create a transparent, traceable, and automated supply chain.
Key technologies:
Blockchain Ledger
Smart Contracts
Automated Compliance Checks
Payment Automation
Expected outcome: A safer, faster, and digitally connected FMCG supply chain.

Summary
Green Haven Foods operates in the FMCG sector, supplying perishable snack products to both urban and rural markets where maintaining product freshness and timely delivery is critical. Existing operations rely heavily on manual processes, resulting in challenges such as inaccurate demand forecasting, delivery delays, inventory shortages, operational inefficiencies, and product waste.
Sales and inventory evaluation identified fluctuating demand patterns, recurring stockouts, and supply–demand mismatches, indicating the need for stronger planning and inventory control mechanisms. Digital integration across procurement, production, warehousing, logistics, and customer delivery was recommended to improve operational coordination and visibility.
AI-driven forecasting methods were proposed to enhance demand prediction, production planning, and transportation cost optimisation through data-based decision making. IoT technologies, including GPS tracking, RFID systems,

environmental sensors, and monitoring dashboards, were recommended to strengthen delivery tracking, warehouse monitoring, inventory accuracy, and freshness management. Blockchain technology was designed to improve supply chain traceability, automate compliance verification, and enable smart-contract-based payment processing after verified deliveries. Sustainability performance was evaluated using Life Cycle Assessment (LCA) concepts aligned with ELCD 3.2 principles, focusing on carbon emissions, water consumption, resource utilisation, waste reduction, and supplier responsiveness.

The recommended digital transformation framework supports improved forecasting accuracy, stronger inventory management, enhanced operational visibility, reduced waste, optimised logistics performance, and improved sustainability outcomes, contributing to a more efficient, transparent, and resilient supply chain system for Green Haven Foods.